Bayside Property Analysis - Assignment 5

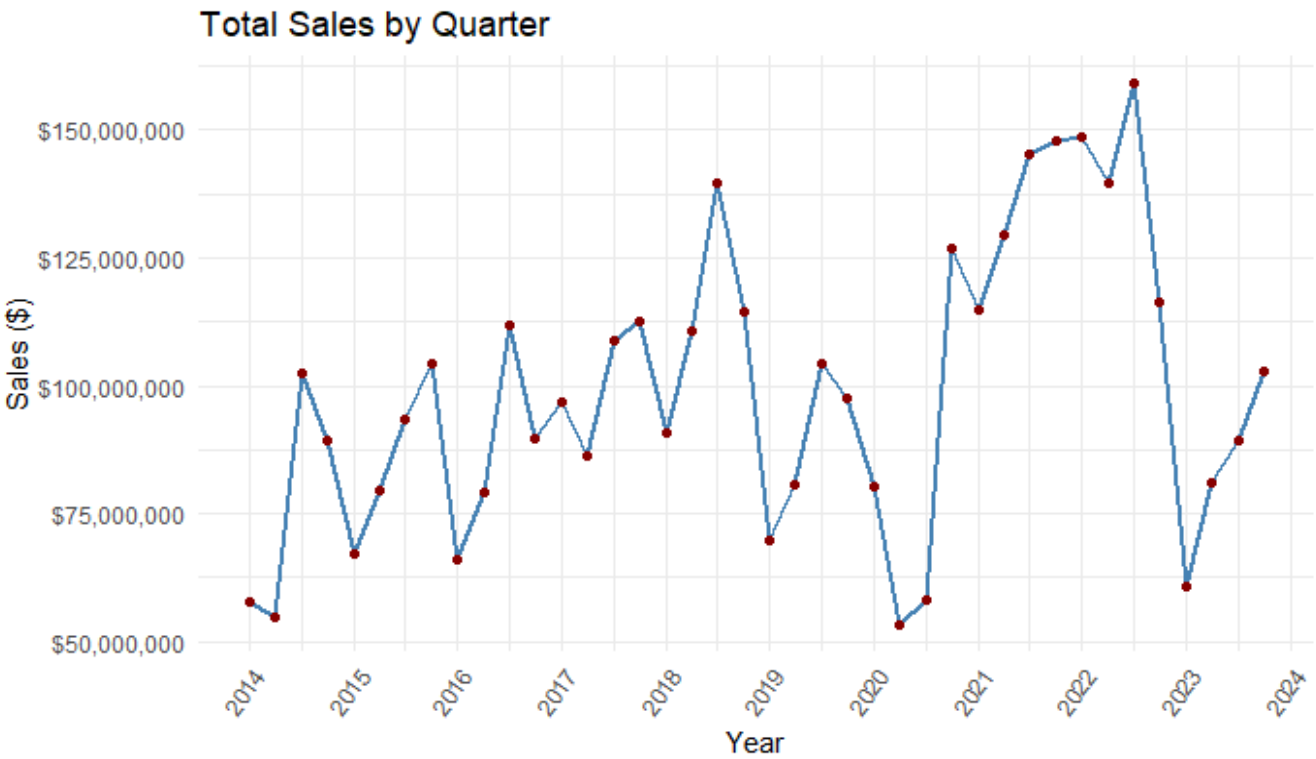
Bowen de Gouw

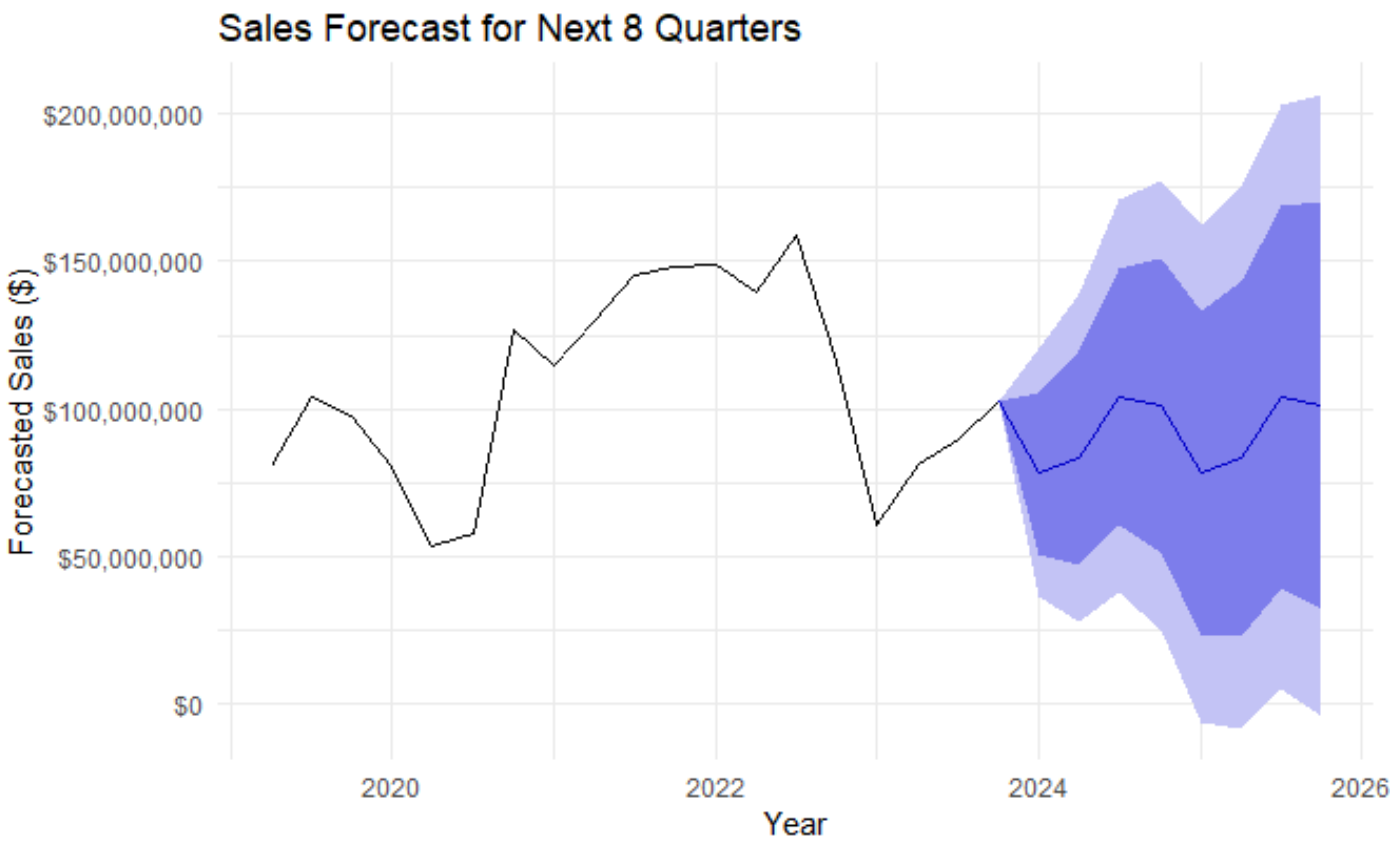
Boston University, Metropolitan College

AD 571

Professor David Ritt

1. **Forecasting**





Year/Quarter	Point.Forecast	Lo.80	Hi.80	Lo.95	Hi.95
2024 Q1	$ 78,227,340.00	$ 50,839,567.67	$ 105,615,112.33	$ 36,341,356.77	$ 120,113,323.23
2024 Q2	$ 83,203,807.97	$ 46,808,492.91	$ 119,599,123.04	$ 27,541,976.91	$ 138,865,639.03
2024 Q3	$ 104,125,675.14	$ 60,546,461.69	$ 147,704,888.58	$ 37,477,019.61	$ 170,774,330.66
2024 Q4	$ 101,011,255.66	$ 51,273,864.88	$ 150,748,646.44	$ 24,944,480.64	$ 177,078,030.69
2025 Q1	$ 78,227,340.00	$ 23,015,593.72	$ 133,439,086.28	-$ 6,211,739.27	$ 162,666,419.27
2025 Q2	$ 83,203,807.97	$ 23,013,561.15	$ 143,394,054.80	-$ 8,849,230.84	$ 175,256,846.78
2025 Q3	$ 104,125,675.14	$ 39,338,371.61	$ 168,912,978.67	$ 5,042,044.80	$ 203,209,305.47
2025 Q4	$ 101,011,255.66	$ 31,931,194.17	$ 170,091,317.16	-$ 4,637,581.43	$ 206,660,092.76

In order to create this forecast, I used the ETS(A,N,A) model because my time series showed clear seasonal patterns for each quarter, but no consistent upward or downward trend. The model includes the following:

- An Additive error which is suitable for data with constant variance
- No trend as the data does not show long-term increase or decrease
- Additive seasonality as it fits repeating quarterly sales patterns

The ETS function in R automatically selected this model based on the data's structure and minimized AIC, which indicated the best fit.

Below is the summary of the model.

```
ETS(A,N,A)

Call:
ets(y = sales_ts)

  Smoothing parameters:
    alpha = 0.8752
    gamma = 1e-04

  Initial states:
    l = 89707762.1609
    s = 9367179 12482988 -8437159 -13413007

  sigma:  21370792

     AIC     AICc      BIC
1505.257 1508.757 1517.079

Training set error measures:
                   ME     RMSE      MAE       MPE
Training set 55251.32 19702897 14683007 -3.155364
                 MAPE      MASE       ACF1
Training set 16.46455 0.5416503 0.01550453
```

2. Optimization

Results:

NPV for Base model:

$	**69,113.60**

NPV for Optimized model:

$	**69,441.15**

Optimized Commission Rate	**5.08%**
New Market Penetration	**2.94%**

Increase in NPV from Base Model to Optimized model	**$**	**327.55**

Screenshots of Excel Sheet:

Base Model

Base Case Assumptions	
Company Comission (per Property Sold)	5.00%
Company Variable Cost (per Property Sold)	1.00%
Company Market Pentration	3.00%
Company Quarterly Fixed Cost	$100,000
Company Nominal Cost of Funds, Compounded Quarterly	10.00%

Year/Quarter	Point.Forecast	Lo.80	Hi.80	Lo.95	Hi.95
2024 Q1	$ 78,227,340.00	$ 50,839,567.67	$ 105,615,112.33	$ 36,341,356.77	$ 120,113,323.23
2024 Q2	$ 83,203,807.97	$ 46,808,492.91	$ 119,599,123.04	$ 27,541,976.91	$ 138,865,639.03
2024 Q3	$ 104,125,675.14	$ 60,546,461.69	$ 147,704,888.58	$ 37,477,019.61	$ 170,774,330.66
2024 Q4	$ 101,011,255.66	$ 51,273,864.88	$ 150,748,646.44	$ 24,944,480.64	$ 177,078,030.69
2025 Q1	$ 78,227,340.00	$ 23,015,593.72	$ 133,439,086.28	-$ 6,211,739.27	$ 162,666,419.27
2025 Q2	$ 83,203,807.97	$ 23,013,561.15	$ 143,394,054.80	-$ 8,849,230.84	$ 175,256,846.78
2025 Q3	$ 104,125,675.14	$ 39,338,371.61	$ 168,912,978.67	$ 5,042,044.80	$ 203,209,305.47
2025 Q4	$ 101,011,255.66	$ 31,931,194.17	$ 170,091,317.16	-$ 4,637,581.43	$ 206,660,092.76

Year/Quarter	Company Market Penetration	Company Commision	Company Variable Cost	Company Fixed Cost	Net Profit
2024 Q1	$ 2,346,820.20	$ 117,341.01	$ 23,468.20	$100,000	-$ 6,127.19
2024 Q2	$ 2,496,114.24	$ 124,805.71	$ 24,961.14	$100,000	-$ 155.43
2024 Q3	$ 3,123,770.25	$ 156,188.51	$ 31,237.70	$100,000	$ 24,950.81
2024 Q4	$ 3,030,337.67	$ 151,516.88	$ 30,303.38	$100,000	$ 21,213.51
2025 Q1	$ 2,346,820.20	$ 117,341.01	$ 23,468.20	$100,000	-$ 6,127.19
2025 Q2	$ 2,496,114.24	$ 124,805.71	$ 24,961.14	$100,000	-$ 155.43
2025 Q3	$ 3,123,770.25	$ 156,188.51	$ 31,237.70	$100,000	$ 24,950.81
2025 Q4	$ 3,030,337.67	$ 151,516.88	$ 30,303.38	$100,000	$ 21,213.51

Total Net Profit	$ 79,763.39
Total NPV	$ 69,113.60

Optimized Model

Base Case Assumptions	
Company Comission (per Property Sold)	5%
Company Variable Cost (per Property Sold)	1.00%
Company Market Pentration	3.00%
Company Quarterly Fixed Cost	$100,000.00
Company Nominal Cost of Funds, Compounded Quarterly	10%
Change in Commission	1.67%
Change in Penetration	-2.00%
New Base Penetration	2.94%
Optimal Comission	5.08%

Year/Quarter	Point.Forecast	Lo.80	Hi.80	Lo.95	Hi.95
2024 Q1	$ 78,227,340.00	$ 50,839,567.67	$ 105,615,112.33	$ 36,341,356.77	$ 120,113,323.23
2024 Q2	$ 83,203,807.97	$ 46,808,492.91	$ 119,599,123.04	$ 27,541,976.91	$ 138,865,639.03
2024 Q3	$ 104,125,675.14	$ 60,546,461.69	$ 147,704,888.58	$ 37,477,019.61	$ 170,774,330.66
2024 Q4	$ 101,011,255.66	$ 51,273,864.88	$ 150,748,646.44	$ 24,944,480.64	$ 177,078,030.69
2025 Q1	$ 78,227,340.00	$ 23,015,593.72	$ 133,439,086.28	-$ 6,211,739.27	$ 162,666,419.27
2025 Q2	$ 83,203,807.97	$ 23,013,561.15	$ 143,394,054.80	-$ 8,849,230.84	$ 175,256,846.78
2025 Q3	$ 104,125,675.14	$ 39,338,371.61	$ 168,912,978.67	$ 5,042,044.80	$ 203,209,305.47
2025 Q4	$ 101,011,255.66	$ 31,931,194.17	$ 170,091,317.16	-$ 4,637,581.43	$ 206,660,092.76

Year/Quarter	Company Market Penetration	Company Commision	Company Variable Cost	Company Fixed Cost	Net Profit
2024 Q1	$ 2,299,883.80	$ 116,910.76	$ 22,998.84	$100,000	-$ 6,088.08
2024 Q2	$ 2,446,191.96	$ 124,348.09	$ 24,461.92	$100,000	-$ 113.83
2024 Q3	$ 3,061,294.85	$ 155,615.82	$ 30,612.95	$100,000	$ 25,002.87
2024 Q4	$ 2,969,730.92	$ 150,961.32	$ 29,697.31	$100,000	$ 21,264.01
2025 Q1	$ 2,299,883.80	$ 116,910.76	$ 22,998.84	$100,000	-$ 6,088.08
2025 Q2	$ 2,446,191.96	$ 124,348.09	$ 24,461.92	$100,000	-$ 113.83
2025 Q3	$ 3,061,294.85	$ 155,615.82	$ 30,612.95	$100,000	$ 25,002.87
2025 Q4	$ 2,969,730.92	$ 150,961.32	$ 29,697.31	$100,000	$ 21,264.01

Total Net Profit	$ 80,129.96
Total NPV	$ 69,441.15
Increase in NPV from Base Model	$ 327.55

Code:

```
library(tidyverse)
library(dplyr)
library(ggplot2)
library(scales)
library(lubridate)
library(factoextra)
library(patchwork)
library(forecast)

BOROUGH <- read_csv("BOROUGH.csv")
BUILDING_CLASS <- read_csv("BUILDING_CLASS.csv")
NEIGHBORHOOD <-read_csv("NEIGHBORHOOD.csv")
NYC_TRANSACTION_DATA <- read_csv("NYC_TRANSACTION_DATA.csv")

all_nyc <- NYC_TRANSACTION_DATA %>% left_join(BUILDING_CLASS, by
=c("BUILDING_CLASS_FINAL_ROLL" = "BUILDING_CODE_ID")) %>%
  left_join(NEIGHBORHOOD, by=c("NEIGHBORHOOD_ID" = "NEIGHBORHOOD_ID"))

bayside <- filter(all_nyc, NEIGHBORHOOD_NAME == "BAYSIDE")

bayside_residential <- bayside %>%
  filter(
    TYPE == "RESIDENTIAL",
    SALE_PRICE >= 10000,
    SALE_PRICE <= 2500000,
    GROSS_SQUARE_FEET > 0,
    GROSS_SQUARE_FEET <= 5000,
  )

bayside_residential <- bayside_residential %>%
  mutate(
    YEAR = year(SALE_DATE),
    QUARTER = quarter(SALE_DATE),
    QDATE = as.Date(paste0(YEAR, "-", (QUARTER - 1) * 3 + 1, "-01"))
  )

recent_quarters <- bayside_residential %>%
  distinct(QDATE)%>%
  arrange(desc(QDATE))%>%
  slice(1:40)

bayside_last40q <- bayside_residential %>%
  semi_join(recent_quarters, by = "QDATE")
```

```
sales_by_quarter <- bayside_last40q %>%
  group_by(QDATE) %>%
  summarise(
    total_sales = sum(SALE_PRICE, na.rm = TRUE)
  ) %>%
  arrange(QDATE)

ggplot(sales_by_quarter, aes(x = QDATE, y = total_sales)) +
  geom_line(color = "steelblue", size = 1) +
  geom_point(color = "darkred") +
  labs(
    title = "Total Sales by Quarter",
    x = "Year",
    y = "Sales ($)"
  ) +
  scale_x_date(date_labels = "%Y",
               date_breaks = "year",
               ) +
    scale_y_continuous(labels = dollar_format()) +
    theme_minimal() +
    theme(axis.text.x = element_text(angle =55, hjust = 1))



start_year <- year(min(sales_by_quarter$QDATE))
start_quarter <- quarter(min(sales_by_quarter$QDATE))

sales_ts <- ts(
sales_by_quarter$total_sales,
start = c(start_year, start_quarter),
frequency = 4
  )

sales_ets <- ets(sales_ts)
sales_forecast <- forecast(sales_ets, h = 8)

autoplot(sales_forecast, include = length(sales_ets)) +
  autolayer(sales_forecast,showgap = FALSE)+
    labs(title = "Sales Forecast for Next 8 Quarters",
         x = "Year",
         y = "Forecasted Sales ($)") +
    scale_y_continuous(labels = scales::dollar_format()) +
    theme_minimal()
```

```
summary(sales_ets)
sales_ets
summary(sales_forecast)
sales_forecast

write.table(sales_forecast, "clipboard", sep = "\t", row.names = TRUE, col.names = TRUE)
```